

# Ioannis Tsioulis

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## Profile

PhD Candidate in Deep Learning (AUTH) · Accredited Lecturer (University of Derby, UK)

PhD candidate in Deep Learning at AUTH, and Accredited Lecturer at Mediterranean College, Athens (School of Computing), an accredited partner of the University of Derby (UK), teaching University of Derby AI and Data Science modules. Research spans applied deep learning, medical AI, deep learning in physics & astronomy, and AI & blockchain.

## Education

**PhD** **Aristotle University of Thessaloniki (AUTH)**, Deep Learning — Thessaloniki, Greece
- In progress.

**MSc** **University of Piraeus**, Cybersecurity and Data Science — Piraeus, Greece, 2023
- Awarded a merit-based scholarship.
- Thesis: Rapid calculation of the signal-to-noise ratio of gravitational-wave sources using artificial neural networks (2023). dione.lib.unipi.gr

**BSc** **National and Kapodistrian University of Athens (NKUA)**, Mathematics — Athens, Greece

## Experience

**Mediterranean College, Athens, School of Computing**, Accredited Lecturer (University of Derby, UK) — Athens, Greece, Oct 2024 – present
- Teaching University of Derby UG and PG modules in Artificial Intelligence and Data Science.
- Supervising BSc and MSc dissertations on applied AI.
- Leading the SuiHub Athens / Mediterranean College applied-research collaboration (AI & blockchain).

## Research Interests

**Areas:** Applied Deep Learning; Medical AI & Computer-Aided Diagnosis; Deep Learning in Physics & Astronomy; AI & Blockchain Applications

## Publications

**ΦasmaNet: A Staged Spectral Token-Refinement Architecture for Fast Gravitational-Wave Signal-to-Noise Ratio Prediction** — 2026
Tsioulis, I., Tefas, A., Stergioulas, N.
arxiv.org/abs/2611.05309 arXiv preprint arXiv:2611.05309

**Human-AI Collaborative Frameworks for Alzheimer's Staging: Evaluating Data Requirements for Augmented Clinical Decision Support** — 2026
Hryb, A., Tsioulis, I., Papadaki, M., Tsoutsi, N.-M., Nassi, M.
HAICAI 2026, Springer LNNS (accepted, in press)

## Conference Presentations & Talks

- Hryb, A., Tsioulis, I., Papadaki, M., Tsoutsi, N.-M., Nassi, M. (2026). Human-AI Collaborative Frameworks for Alzheimer's Staging: Evaluating Data Requirements for Augmented Clinical Decision Support. Oral presentation, HAICAI 2026, University of West Attica, Athens. Conference programme
- Tsioulis, I., Hryb, A., Papadaki, M. (2026). Ταξινόμηση Σταδίων Νόσου Alzheimer από MRI με CNN και Εκτίμηση Απαιτήσεων Δεδομένων. Oral presentation, 52ο Ετήσιο Πανελλήνιο Ιατρικό Συνέδριο, Athens. Book of abstracts, p. 80
- Tsioulis, I. (2023). Rapid calculation of the signal-to-noise ratio of gravitational-wave sources using artificial neural networks. Oral presentation, 20th Conference on Recent Developments in Gravity (NEB-20, HSRGC), Eugenides Foundation, Athens. Contribution page
- Tsioulis, I. (2025). AI in Pharmacovigilance: the Intersection of AI and Drug Safety. Guest lecture, School of Medicine, NKUA.

## Datasets

- Tsioulis, I., Chaikalis, T., Anastasopoulos, K., Chanoutsos, V. E., Chrysomallis, N., Gubochkin, D. V., Margetis, A., Patrakis, V., Trousas, N., Tsipis, G., Zerkidis, D. (2026). Sui Move Issues: A Labeled Dataset for Smart Contract Error Classification (v1). Zenodo. doi.org/10.5281/zenodo.19682589. Supervised by I. Tsioulis; produced with BSc/MSc students at Mediterranean College in collaboration with SuiHub Athens.

## Software

**PhasmaNet: a deep learning library for tabular data**
Open-source Python package of the ΦasmaNet architecture (pip install phasmanet), with A. Tefas and N. Stergioulas. PyPI, GitHub, Docs, Paper

## Projects

**NiceMove: AI-Powered Move Smart-Contract Linter**
VS Code extension that flags Sui Move smart-contract issues with a small local model.
VS Code Marketplace

- Research project I lead with SuiHub Athens / Mysten Labs; 11 BSc/MSc students under my supervision.
- Open dataset of 3,321 labelled Sui Move snippets, with 10 student co-authors. Zenodo
- Model and extension finalised in an MSc thesis (V. Bakalis): ~99% preliminary accuracy, above general-purpose LLMs.

## Selected Supervision

**MSc — Vasileios Bakalis (2026):** *LLM-Based Classification of Move Smart-Contract Issues in the Sui Ecosystem: Dataset Consolidation and Baseline Model Comparison.* MSc in Big Data Analytics, Mediterranean College / University of Derby. Built the curated dataset of ~3,300 labelled Sui Move snippets underpinning NiceMove. PDF

**BSc — Vasileios Eirinaios Chanoutsos (2026):** *Security Comparison of Smart Contracts in Sui vs. Other Blockchains.* BSc (Hons) Computer Science, Mediterranean College / University of Derby. PDF

**BSc — Dimitrios Zerkidis (2026):** *AI-Based Code Style Enforcement with Code Llama.* BSc (Hons) Computer Science, Mediterranean College / University of Derby. PDF

## Awards

- Merit-based Scholarship, MSc in Cybersecurity and Data Science, University of Piraeus.

## Skills

**Languages:** Python, SQL, C

**ML / DL:** PyTorch, deep learning, CNNs, transformers, continual learning

**Domains:** Applied Deep Learning

## Military Service

**Hellenic Army (2017):** Presidential Guard (Evzones), Athens. Compulsory national service, completed